



Alberta Public Health Disease Management Guidelines

Respiratory Syncytial Virus



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For further information on the use of this guideline contact:

Health.CD@gov.ab.ca

Communicable Disease Control Branch

Public Health Division

Alberta Health

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Revisions

Revision Date	Document Section	Description of Revision
August 2024	General	Entire guideline revised to align with current information
	Reporting Requirements	Revised reporting requirements
	Clinical Assessment and Epidemiology	New
	Public Health Management	New

Case Definition

Confirmed Case

Clinical illness ^(A) with laboratory confirmation of infection:

- Detection of respiratory syncytial virus (RSV) nucleic acid (e.g., polymerase chain reaction [PCR]) in an appropriate clinical specimen (e.g., nasopharyngeal swab or aspirate, throat swab^(B), auger suction, bronchoalveolar lavage [BAL], or bronchoalveolar wash)^(C);

OR

- Detection of RSV viral antigen in nasopharyngeal cells by Immunofluorescence Assay (IFA) or Enzyme Immunoassay (EIA);

OR

- Isolation of RSV from respiratory secretions in cell culture^(D).

Probable Case (Outbreak Only)

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case.

^(A) Primary infection manifests as pneumonia, bronchiolitis, tracheobronchitis or upper respiratory tract illness (often accompanied by fever and otitis media). The infection is very rarely asymptomatic.

^(B) Although throat swabs are not typically used in RSV diagnostic testing, a positive throat specimen would still be considered laboratory confirmation of RSV infection.

^(C) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

^(D) Although culture is not routinely used for RSV diagnosis in Alberta, a positive RSV culture result would still be considered laboratory confirmation of RSV infection⁽⁸⁾.

Reporting Requirements

Laboratories

- All laboratories shall report all positive RSV laboratory results by mail, fax, or electronic transfer within 48 hours (two business days) to the Medical Officer of Health (MOH) (or designate) of the zone.
- All laboratories shall report all (positive and negative) RSV laboratory results by mail, fax, or electronic transfer within 48 hours (two business days) to the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall report all confirmed RSV hospitalizations/deaths ^(E) to the CMOH (or designate) via the [COVID-19/Seasonal Influenza/RSV death and hospitalized case report form](#) using existing processes (e.g., CDOM or confidential fax).
 - The report form must be submitted within **one week** of notification of hospitalization, discharge from hospital, resolution of the RSV case status or death.^(E)
- The MOH (or designate) shall report all RSV outbreaks to the CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or confidential fax).
 - The initial report form must be submitted within **24 hours** of opening the outbreak investigation and the final report must be submitted within **48 hours** of closing the outbreak investigation.
 - Aggregate numbers must be reported weekly for outbreaks in continuing care and acute care settings while they remain open.
- For out-of-province and out-of-country confirmed RSV cases that are reported as hospitalized/deceased, the MOH (or designate) of the zone where the case resides shall forward the following information to the CMOH (or designate) using existing processes (e.g., CDOM) within **48 hours** (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (if applicable), and
 - other relevant clinical/epidemiological information.

^(E) Confirmed RSV deaths: deaths where a lab-confirmed RSV infection is the direct or contributing cause that come to the attention of Public Health, including deaths in the community.

Clinical Assessment and Epidemiology

Etiology

RSV is an RNA virus of the *Orthopneumovirus* genus and the *Pneumoviridae* family.^(1,2) Human RSV has two antigenic subgroups, A and B, with both often circulating during the same RSV season.⁽²⁾

Clinical Presentation

RSV is a common respiratory virus that typically causes mild cold-like symptoms that resolve within 1 to 2 weeks.^(3,4) Symptoms usually appear in stages and include rhinorrhea, coughing, sneezing, headache, fatigue, decrease in appetite, and pharyngitis.^(4,5) If bronchiolitis occurs, additional symptoms include wheezing, tachypnea, intercostal retractions, grunting and nasal flaring; fever may or may not be present.⁽²⁾

RSV is common in childhood and is the most common cause of lower respiratory tract illness in young children.^(5,6) Although RSV most commonly presents as an upper respiratory tract infection, approximately 20-30% of infants develop lower respiratory tract illness such as bronchiolitis, tracheobronchitis, or pneumonia with the first infection.^(2,5) Infants with RSV may present with non-specific signs and symptoms such as lethargy, difficulty breathing, difficulty feeding, and/or irritability, tachypnea, and cyanosis.^(1,5) Infants who develop RSV infection in the first weeks of life, especially preterm infants, are at high risk of developing apnea.⁽²⁾ Children under 5 may experience rapid breathing, trouble swallowing, or sepsis.⁽⁷⁾

Healthy adults and infants infected with RSV usually do not require hospitalization. Some older adults and/or infants younger than 6 months of age may require hospitalization if they are having trouble breathing or are dehydrated; however, in most cases hospitalization lasts a few days.⁽⁴⁾ Approximately 1-3% of all children under age 1 will be hospitalized because of lower respiratory tract infections caused by severe RSV, with the highest rate of RSV hospitalization occurring in the first six months of life.⁽²⁾

Diagnosis

Diagnosis of RSV is generally made through nucleic acid amplification tests (e.g., RT-PCR, LAMP).^(2,8) Other diagnostic assays such as direct fluorescent antibody (DFA) and enzyme or chromatographic immunoassays that detect viral antigen in nasopharyngeal specimens have lower sensitivity.^(2,8)

Treatment

Treatment with antiviral medication is not routinely recommended and most infections resolve within 1 to 2 weeks.⁽⁴⁾ Supportive care is recommended for individuals with mild to moderate RSV infection.⁽⁵⁾ Treatment for severe illness, such as bronchiolitis in hospitalized infants, may include hydration, airway suction, and/or continuous positive airway pressure (CPAP).⁽²⁾

Reservoir

Humans.^(1,2)

Transmission

RSV is spread through particles and droplets released into the air when an infected person coughs, sneezes, breathes or speaks, or via other close contact such as kissing.⁽⁷⁾ Indirect transmission may also occur by touching surfaces contaminated with RSV and then touching eyes, nose or mouth.^(2,7) RSV can survive on surfaces for several hours and on hands for 30 minutes or more.⁽²⁾

Incubation Period

The incubation period for RSV ranges from two to eight days, with four to six days being the most common.⁽²⁾

Period of Communicability

The period of communicability for RSV can begin one to two days prior to symptom onset. Viral shedding is typically three to eight days.^(2,4) In young infants and immunosuppressed children, viral shedding may last three to four weeks or longer.⁽²⁾

Host Susceptibility

Susceptibility is universal. Almost all children are infected by RSV at least once by age two and may experience reinfections throughout life.^(2,7) Subsequent infections can occur at any age as infection produces only partial and temporal immunity.⁽⁵⁾ However, subsequent infections are usually less severe.⁽²⁾

The risk of severe outcomes from RSV is higher among infants and children under two years of age, adults and children with chronic lung disease, congenital heart disease or compromised immune systems, and adults over 65.⁽⁵⁾

Incidence

RSV infection is a seasonal illness; however, the onset, duration and peak varies annually. In the northern hemisphere, it typically lasts from November to April, however it can be seen year-round in tropical and equatorial regions.⁽¹⁾

For more information on global RSV surveillance, refer to the [World Health Organization \(WHO\) RSV Surveillance Strategy](#).

For more information on seasonal RSV activity in Canada, refer to the [Respiratory Virus Detection Surveillance](#) website.

In Alberta, hospitalizations/deaths where a lab-confirmed RSV infection is the direct or contributing cause became reportable in 2024. Alberta RSV data can be found on the [Alberta respiratory virus dashboard](#) website.

Public Health Management (Outbreak Only)

Key Investigation

- Refer to the Management of RSV [Outbreaks in Congregate Settings section](#).
- RSV outbreaks should be managed as per direction from the Zone MOH.
- The Zone MOH will determine the need and extent of outbreak control measures.
- For additional outbreak management measures in different settings refer to [AHS guides for outbreak prevention and control](#).

Management of a Case

- Individuals with symptoms of respiratory illness should stay home. If symptoms worsen, they should seek medical guidance or care.
- General guidance in high-risk settings should include infection prevention and control precautions to prevent disease transmission.

Management of Contacts

- Individual management of close contacts is NOT required and testing through the public health system is NOT indicated for close contacts.

Management of RSV Outbreaks in Congregate Settings

Table 1: RSV Outbreak Definitions

Type of Setting	Example	Confirmed RSV Outbreak**
Congregate ^(F) Care Facility	Licensed supportive living (including lodges), long-term care (nursing homes and auxiliary hospitals), and hospices	2 confirmed cases in residents^Ω
Congregate ^(F) Living Settings	Prisons/Correctional Facilities	2 confirmed cases in residents^Ω
Other Settings	MOHs can exercise their authority for further investigation in any situation with unusual number of illness^(G)	

**Confirmed resident/patient case(s) needs to have been in the setting during their incubation period or infectious period. For more information refer to the [AHS Outbreak Prevention and Control Guides](#).

^ΩCases within an 8-day period with an epidemiological link (i.e., an exposure at a common setting, or time spent in a common location or venue, where there is reasonable evidence that transmission could have occurred

NOTE: RSV outbreaks can be closed 8 days after onset of symptoms in the last patient/resident case.

Table 2: Mixed Pathogen Outbreak Classification in Congregate Care/Living Facility

Cases Identified	Outbreak Classification	Outbreak Closed
≥2 confirmed RSV cases with an epi-link AND ≤1 confirmed case of any other respiratory pathogen (e.g., influenza, COVID-19)	RSV	8 days after onset of symptoms in the last RSV case
≥2 confirmed RSV cases with an epi-link AND ≥2 confirmed influenza cases with an epi link	RSV/influenza	8 days after onset of symptoms in the last case
≥2 confirmed RSV cases with an epi-link AND ≥2 confirmed COVID-19 cases with an epi-link	RSV/COVID-19	14 days after onset of symptoms in the last case*
≥2 ILI cases with an epi-link AND ≤1 confirmed case of any other respiratory pathogen (e.g., influenza, RSV, COVID-19)	ILI	Based on the organism identified (if applicable)

***NOTE:** Outbreaks can be closed 14 days after symptom onset of the last patient/resident case regardless if additional staff cases are found at the tail end of an outbreak

^(F) Congregate settings are defined as locations where individuals live, work or are cared for within close quarters in a communal environment.

^(G) For information on respiratory illness outbreaks in schools, childcare settings, and homeless shelters/temporary housing, please see [Appendix: Management of Respiratory Illness Outbreaks](#).

Preventative Measures

- Prevention is the most effective strategy to control RSV infections as treatment options are limited.
- Provide general and ongoing public education regarding preventative measures against respiratory illnesses including RSV such as staying away from people who are sick, staying home if sick, covering coughs and sneezes (respiratory etiquette), practicing frequent hand hygiene, and wearing a mask for 10 days after respiratory illness onset, while indoors with other people.
- Some infants may be eligible to receive RSV immunoprophylaxis. Refer to the [Alberta Pediatric RSV Prevention Program](#) for more information.
- For additional information on vaccines available in Alberta, refer to the [Alberta Immunization Policy \(AIP\)](#).

Appendix: Management of Respiratory Illness Outbreaks

- In schools, child care settings and homeless shelters/temporary housing, the [Respiratory Illness \(RI\) definition](#) and the [Respiratory Illness Outbreak \(RIO\) definition](#) will be used to identify, report and manage outbreaks that may be caused by a variety of respiratory pathogens, including RSV.

Respiratory Illness (RI) Definition:

- NEW onset of TWO or more symptoms and at least one must be from List A:
 - List A:** cough, shortness of breath (SOB), sore throat, loss or altered sense of taste/smell, runny nose/nasal congestion
 - List B:** fever, fatigue (significant and unusual), muscle ache/joint pain, headache, nausea/diarrhea

Respiratory Illness Outbreak (RIO) Definition

- Two or more cases with respiratory illness within **7 days** with a common epidemiological link

AND

- NO respiratory pathogen identified OR one case of any respiratory pathogen identified (e.g., Influenza; COVID-19; RSV)

Table 3: RIO Definitions

Setting	Confirmed Respiratory Illness Outbreak
Schools (K-12)	10% absenteeism due to respiratory illness OR an unusual number of students with similar respiratory symptoms AND at least two epi-linked individuals within the school setting (who are not from the same household) with respiratory illness symptom onset within a 7-day period.
Childcare settings: Daycares, after school care, preschools	At least two epi-linked cases with respiratory illness in individuals in the childcare cohort (not from the same household) with symptom onset within a 7-day period.
Homeless shelters/temporary housing	Unusual number of clients (above baseline) with respiratory illness and at least two epi-linked individuals within the setting with respiratory illness symptom onset within a 7-day period.

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